

SGEIA — RAG Pipeline

Smart Grid Energy Intelligence Assistant

OFFLINE — Build Knowledge Base (run once at setup)

DS1 — Grid Stability Dataset

60,000 rows · tau, p, g features · stabf label

DS2 — Smart Meter Dataset

2,000,000 rows · voltage · current · sub-metering

200 Synthetic Incidents CSV

Generated from DS1 + DS2 · The RAG knowledge base

CHUNKING

1 incident = 1 chunk · 40-80 words · No splitting

EMBEDDING

all-MiniLM-L6-v2 · 384-dim vector · Runs on CPU offline

BM25 INDEX

Keyword inverted index · Exact matches

CHROMADB

Vector store · Cosine similarity · Persists to disk

USER QUESTION

Types in plain English · e.g. 'Why is Zone A unstable?'

ONLINE — Every User Question (real-time, ~)

INPUT GUARDRAILS

Format · Harmful content · Domain check · PII masking

PASS

CACHE LOOKUP

Same question before? cosine > 0.97 → instant reply!

HIT →

MISS

ORCHESTRATOR AGENT

Classifies intent · Routes to the right agents only

BM25 SEARCH

Finds exact: Zone_A, INC-042

SEMANTIC SEARCH

Finds similar meaning in ChromaDB

ML ANALYSIS

XGBoost + Isolation Forest

RRF FUSION

Combines BM25 + Semantic · Picks Top 8 best incidents

LLM PROCESSING

Groq llama · Reads question + 8 incidents + ML stats → writes answer

instant
reply

RECOMMENDATION GENERATION

Root cause · Mitigation steps · Resolution suggested

DEEPEVAL QUALITY

Faithfulness % · LLM Judge /5 · Output Safety · Retry if fail

PASS

OUTPUT GUARDRAILS

Jailbreak scan · PII re-scan on output · Length check

ANSWER TO OPERATOR

Markdown + DeepEval scores shown + feedback

Colour key:

■ Data/Index

■ Search

■ Quality

■ Safety

■ feedback

■ Cache

■ ML Models

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